

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2209

Denton West Interchange 100 kV
May 17, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2209. The Interconnection Customer proposes a 229.9 MW solar photovoltaic facility interconnecting at the Denton West Interchange 100 kV substation in the Illinois area. The request was received on October 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 572 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$596,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
RIVERTEX SUBSTATION / DENTON W INTERCHANGE 100.0 kV #2	RIVERTEX / DENTON WEST INTERCHANGE 100.0 kV	112.3	149	7.4
DENTON WEST INTERCHANGE / RIVERTEX 100.0 kV Ckt 1	RIERTEX / DENTON WEST INTERCHANGE 100.0 kV	101.1	166	7.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DENTON WEST INTERCHANGE / RIVERTEX 100.0 kV Line 2	Rebuild line with higher-capacity conductor	\$385,000	100% (\$385,000)
DENTON WEST INTERCHANGE SUBSTATION / RIVERTEX 100.0 kV L	Reconductor line	\$211,000	100% (\$211,000)
	Total allocated to project		\$596,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.